EXPERIMENT-6

BINARY TO GRAY CONVERSION

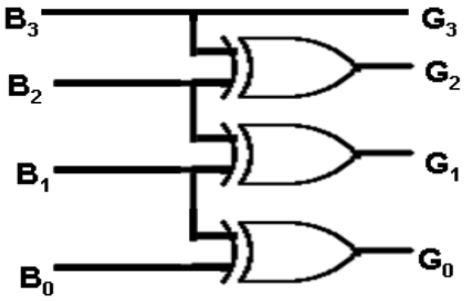
Aim: - To convert binary numbers to gray codes.

EQUIPMENTS / COMPONENTS REQUIRED :- Digital IC Trainer Kit, and IC 7486

Theory

A given binary number can be converted into its Gray code equivalent by going through the following steps:

1. Begin with the most significant bit (MSB) of the binary number. The MSB of the Gray code equivalent is the same as the MSB of the given binary number.
2. The second most significant bit, adjacent to the MSB, in the Gray code number is obtained by adding the MSB and the second MSB of the binary number and ignoring the carry, if any. That is, if the MSB and the bit adjacent to it are both '1', then the corresponding Gray code bit would be a '0'.
3. The third most significant bit, adjacent to the second MSB, in the Gray code number is obtained by adding the second MSB and the third MSB in the binary number and ignoring the carry, if any.
3. The process continues until we obtain the LSB of the Gray code number by the addition of the LSB and the next higher adjacent bit of the binary number.
4.
5. $B_3 = G_3$
6. $B_2 \oplus B_3 = G_2$
7. $B_1 \oplus B_2 = G_1$
8. $B_0 \oplus B_1 = G_0$



Inputs				Outputs			
B3	B2	B1	B0	G3 (V)	G2 (V)	G1 (V)	G0 (V)
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1
0	0	1	0	0	0	1	1
0	0	1	1	0	0	1	0
0	1	0	0	0	1	1	0
0	1	0	1	0	1	1	1
0	1	1	0	0	1	0	1
0	1	1	1	0	1	0	0
1	0	0	0	1	1	0	0
1	0	0	1	1	1	0	1
1	0	1	0	1	1	1	1
1	0	1	1	1	1	1	0
1	1	0	0	1	0	1	0
1	1	0	1	1	0	1	1
1	1	1	0	1	0	0	1
1	1	1	1	1	0	0	0

GRAY TO BINARY CONVERSION

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The Most Significant Bit (MSB) of the binary code is always equal to the MSB of the given binary number.

2.Other bits of the output binary code can be obtained by checking gray code bit at that index. If current gray code bit is 0, then copy previous binary code bit, else copy invert of previous binary code bit.

There are four inputs and four outputs. The input variable are defined as G_3, G_2, G_1, G_0 and the output variables are defined as B_3, B_2, B_1, B_0. From the truth table, combinational circuit is designed.The logical expressions are defined as :

$$G_0 \oplus G_1 \oplus G_2 \oplus G_3 = B_0$$

$$G_1 \oplus G_2 \oplus G_3 = B_1$$

$$G_2 \oplus G_3 = B_2$$

$$G_3 = B3$$

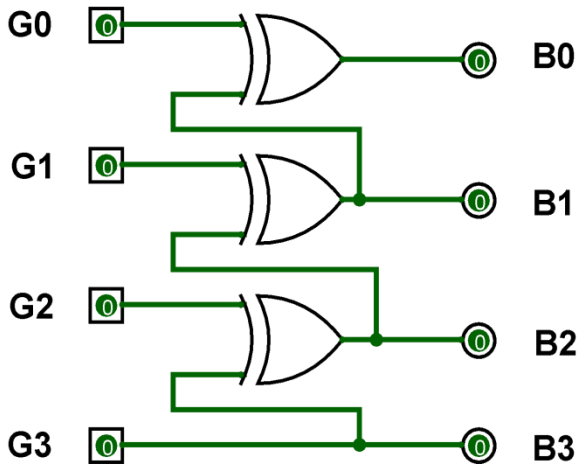


Figure-3: Gray to Binary Code Converter Circuit

Gray code				Natural-binary code			
G3	G2	G1	G0	B3	B2	B1	B0
0	0	0	0	0	0	0	0
0	0	0	1	0	0	0	1
0	0	1	0	0	0	1	1
0	0	1	1	0	0	1	0
0	1	0	0	0	1	1	1
0	1	0	1	0	1	1	0
0	1	1	0	0	1	0	0
0	1	1	1	0	1	0	1
1	0	0	0	1	1	1	1
1	0	0	1	1	1	1	0
1	0	1	0	1	1	0	0
1	0	1	1	1	1	0	1
1	1	0	0	1	0	0	0
1	1	0	1	1	0	0	1
1	1	1	0	1	0	1	1
1	1	1	1	1	0	1	0

Figure-4: Gray to Binary Code Converter Truth Table